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Display Interfaces

Let the connectors and cables confuse you no more.

Samir Makwana

You've been hearing a lot about High Definition content for quite some time. Everything's gone digital—movies, gaming, TVs, monitors, projectors. With this, a slew of new connector interfaces has arisen, and a lot of us are left scratching our heads wondering what goes where. We're going to do a little show and tell here to help you out!

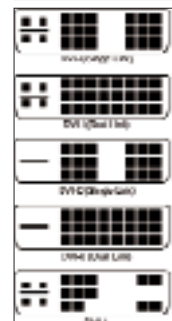
DVI—Digital Video Interface

DVI is widely used for flat-panel LCD monitors, digital projectors, and present-generation graphics cards as a high-resolution digital display interface. The technology was created by the Digital Display Working Group. All video cards produce digital signals that are converted to analogue at the VGA port; they are then transmitted to a monitor where it's re-converted to digital. Instead, display devices that have DVI ports can be connected to receive a direct digital display from the graphics card.

It's not that simple, though, because DVI connectors come in three flavours—DVI-Digital (DVI-D), DVI-Analogue (DVI-A) and DVI-Integrated (DVI-I). The DVI-D connector is used to connect devices

such as digital LCD monitors—it eliminates the analogue signal conversion that takes place at the VGA port of the video card. To carry a digital signal to an analogue display, DVI-A cables convert the signal to analogue and hook into the VGA port of the analogue display. DVI-I cables can be used to transmit both digital-to-digital and analogue-to-analogue signals on different pins.

DVI-D and DVI-I connectors are sub-divided into single- and dual-link connectors—dual-link connectors support 24 bits per pixel and higher resolutions. A single-link DVI connector that consists of four twisted-pair wires (red, blue, green, and one for clock) that offer a maximum display resolution of 1600 x 1200 at 60 Hz. Dual-link DVI connectors offer double the power of transmission and can display a resolution of 2048 x 1536 at 60 Hz. Dual-link connectors are easily identified by their additional pins.



The pin layout differs for each type of DVI Connector

S-Video

This is a high-quality video interface format that carries three-colour YUV analogue video signals—Y (Brightness/Luminance) lies in one channel, U (colour) and V (chroma) in another channel. Since television

sets display Luminance

and Chroma signals separately, sharper images can be displayed using an S-video connector than a Composite video connector.

S-video is capable of a display resolution of 480i and 576i (standard definition), and doesn't carry audio signals on the same cable. Devices such as VCRs, DVD players, camcorders, receivers, and game systems may use 4-pin Mini-DIN (Deutsches Institut für Normung) cables, for which a special S-video connector is needed. For computer connectivity, 7-pin Quasi-DIN S-video connectors are used. Connect S-video with care; the pins are weak and may bend.

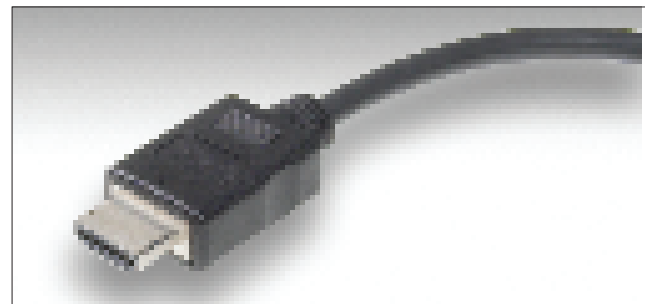


HDMI

High Definition Multimedia Interface is an industry-supported, uncompressed, all-digital audio/video interface. Transferring uncompressed audio/video data over a single cable requires a lot of bandwidth. HDMI is the solution to this, and is capable of transmitting HD content at 5 Gbps (gigabits per second), which is twice the bandwidth needed to transmit multi-channel audio and video.

It is backward-compatible with DVI, and a DVI/HDMI cable can be used to connect your graphics card to, say, an LCD TV. HDMI is specifically used for DRM-protected digital audio/video sources such as gaming consoles, Blu-ray Disc players, etc.

High-bandwidth Digital Content Protection (HDCP) technology developed by Digital Content Protection, LLC, is used



HDMI: the slim connector for fat piping—as in HD content

for protection of content via the HDMI as well as DVI interfaces.

Two types of HDMI connectors are currently available—standard and high-speed. Standard cables can handle 1080i signals (frame resolution) at a speed of 75 MHz, while high-speed cables can handle 1080p signals at speeds of 340 MHz—the highest currently available.

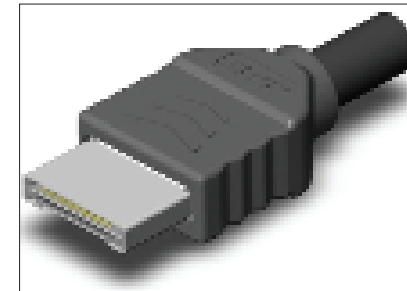
The Standard, also called the “Type A” connector, has 19 pins that provide bandwidth support to SDTV, EDTV, HDTV modes and more. It is electrically compatible with single-link DVI-D, and its width/height is 13.9 by 4.45 mm. The high-speed, also called “Type B” connector, has 29 pins, and is electrically compatible with dual-link DVI-D; its width is 21.2 mm. The maximum length for an HDMI cable is 50 feet. They're expensive, as are DVI to HDMI convertors.

Unified Display Interface

This is one of the latest video interfaces used to connect PCs to

HDTV displays. It will replace VGA connectors on PC monitors as well as other devices and make it compatible with DVI. To make UDI compatible with newer interfaces like HDMI, the HDCP technology was incorporated into it.

Despite having 26 pins, the UDI connector is small, and is cheap to manufacture. This display interface is competing directly with the DisplayPort interface. The future looks bleak though, because its Special Interest Group



UDI: The soon-to-be VGA killer

is on shaky ground after Samsung and Intel exited it this year. Apple currently heads the group.

Display Port

This is a VESA (Video Electronics Standards Association) standardised technology. It is being pitched as a VGA and DVI standards replacement. Though the DisplayPort specifications are similar to HDMI, it will be used primarily as a computer interface—connecting PCs to TVs, CRT monitors, flat panel displays, digital projectors, home entertainment receivers, etc.

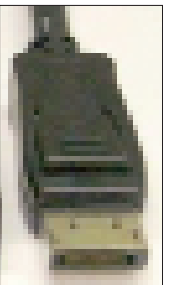
DisplayPort provides four channels of data flow at up to 10.8 Gbps. Device control instructions are carried by a separate bi-directional channel. The DisplayPort connector is screwless and small, designed to offer a maximum length of 15 metres.

Unlike DVI, DisplayPort will be able to carry audio data (up to eight channels) on the same (video) cable. For content protection, DisplayPort Content Protection (DPCP) has been licensed by Philips; it has advanced encryption-based authentication, and is also HDCP-capable.

DisplayPort supports a maximum resolution of 2560 x 2048 at 60 Hz. By 2009, the specifications of DisplayPort will be upgraded to support higher display resolutions like 3840 x 2400 at 60 Hz. According to VESA, DisplayPort is capable of six 1080i streams or three 1080p streams over a single connector. This means you can run multiple displays off a single DisplayPort connector.

DisplayPort is currently compatible with DVI, for which you need to buy DVI-to-DisplayPort cables or adaptors. Likewise, for HDMI, HDMI-to-DisplayPort adaptors are available. ■

samir_makwana@thinkdigit.com



DisplayPort—will it replace both VGA and DVI?

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